VoiceOps 2026: ZentroMart Support Agent

Capstone report · Mithil · github.com/mithilg25/VoiceOps

1. Introduction

This project delivers a grounded, conversational AI support agent for the fictional retailer ZentroMart. The agent handles inquiries regarding orders, returns, refunds, and cancellations, utilizing a RAG pipeline, stateful tool-calling, and Gnani voice integration. The LLM used for inference was *openai/gpt-oss-120b*, hosted via the Groq API.

2. Steps 1 & 2: Building the RAG Pipeline & Evaluation

The Baseline (Ungrounded)

The first full evaluation run scored 23.1% (12/52). During this run, a retrieval bug (detailed in Section 5) caused the vector search to query empty strings, returning irrelevant context. Therefore, this score approximates an ungrounded baseline where the LLM had to guess.

The RAG Implementation

To ground the agent, the 21 Markdown documents were chunked by their *[SECTION-ID]* headers, preserving metadata (status, *effective_from*). Chunks were embedded using *paraphrase-multilingual-MiniLM-L12-v2* to handle Hinglish. The agent retrieves the top k=6 chunks, applying a ranking multiplier so current policies outrank FAQs. The system prompt instructs the model to answer only from the retrieved context.

Evaluation Summary

Evaluation Mode	Score	Notes
Baseline (Ungrounded)	23.1% (12/52)	Estimated. Retrieval bug caused the agent to guess.
Step 2: RAG (Text)	69.2% (36/52)	Successfully retrieved and applied policy chunks.
Step 3: RAG (Convs)	60.9% (14/23)	Stateful multi-turn handling and tool calling.
Step 4: RAG (ASR)	57.7% (30/52)	Evaluated on simulated speech-to-text transcripts.

Retrieval quality. In a manual review of 10 randomly selected questions, the correct *[SECTION-ID]* (matching the dataset's *gold_evidence*) appeared in the top-6 retrieved chunks in 9 of 10 cases. This is a small sample, not a full-set measurement.

3. Step 3: Conversation Handling & Tools

A stateful tool-calling loop was implemented for multi-turn conversations.

- **Identity verification:** the agent asks for the Order ID and one verification value (phone last-4 or pincode). The dispatcher enforces a maximum of 1 retry; if both attempts fail, the agent refuses to share order details.
- **Authoritative math:** the LLM is instructed to use *add_business_days()* and *business_days_between()*, which account for Sundays and warehouse holidays.
- **"I don't know" rule:** for unanswerable questions, the system prompt instructs the agent to state it does not have the information and offer a human handover.
- **Escalation:** the agent is instructed to call *create_ticket()* if the customer asks for a human, mentions legal action, has 2+ prior contacts in 7 days, or reports a damage claim above ₹10,000.

Failure analysis

Of the 23 conversations, 9 failed. Reviewing the logs revealed three patterns:

Failures	Pattern	Detail
4	Dropped context	The agent called *create_ticket* without passing the *order_id* from earlier turns.
3	Missing facts	The agent omitted the full year in dates ("27th" instead of "27 October 2026"), failing automatic grading.
2	Formatting	The agent gave a correct conversational answer but missed the exact required keyword string.

4. Step 4: Voice Integration

The agent was wrapped with Gnani's speech models.

- **Batch processing:** *run.py* accepts a folder of audio files and saves the transcribed text, the synthesized *reply.wav*, and a JSON file containing the retrieved sections.
- **Real audio testing:** recordings of my own voice were used to test Gnani STT/TTS. Gnani handled clear English well, though an empty transcript was occasionally returned when background noise was present. The wrapper then replies, "Sorry, I didn't catch that. Could you say it again?"
- **TTS:** the *timbre-v2.5* model with the "Kaveri" voice is used. A *speechify()* function groups 8-digit order IDs into blocks of four (e.g., "6952 0867").
- **ASR evaluation:** the ASR score (57.7%) was measured on the dataset's *asr_text* column, which is simulated transcript text, not Gnani STT output. The drop from 69.2% to 57.7% indicates the agent likely struggled to map spelled-out SKUs (e.g., "ele-zero zero one") to the correct catalog entries.

5. What Went Wrong (and How It Was Fixed)

- **The silent retrieval bug:** the baseline score was stuck at 23.1%. *Agent.respond()* retrieved documents before appending the user's message to the history, so the RAG search queried an empty string. Reordering the code fixed the pipeline, and the score rose to 69.2%.
- **Null tool arguments:** the LLM passed *null* for optional tool arguments (like *order_id*), causing API validation errors. Fixed by updating the JSON schema to allow *["string", "null"]*.
- **Rate limits:** free API tiers restricted the 200-item evaluation (52 text, 52 ASR, 96 conversation turns). A fixed 15-second retry delay (*time.sleep(15)*) was added to the LLM wrapper to handle HTTP 429 errors.

6. What Would Be Fixed Next

- **Auto-fill tool arguments:** the dispatcher should pass the verified *order_id* into *create_ticket* from the conversation state, removing the cause of the 4 dropped handovers.
- **Format forcing in agent code:** a post-processing step should always include the full year in every date (e.g., "27 October 2026" instead of "the 27th").
- **Fuzzy matching for ASR:** add a layer to correct misspelled SKUs (e.g., "ele zero one" to "ELE-0001") before the LLM processes the text.
- **True Step 1 baseline:** run the evaluation questions with retrieval explicitly disabled in code to get a clean, non-estimated baseline.
- **Full retrieval measurement:** compute recall against *gold_evidence* across all answerable questions instead of a 10-question sample.

7. Bonus Work

Beyond the base requirements, two features were added:

- **Multilingual embeddings:** switched to *paraphrase-multilingual-MiniLM-L12-v2* to better support Hinglish queries.
- **Authoritative date tools:** dedicated Python tools for business-day math, so refund timelines do not depend on the LLM's own date arithmetic.

8. Conclusion

The RAG pipeline raises practice-question accuracy from an estimated 23.1% to 69.2%. The agent handles a majority of the practice conversations (60.9%) and 57.7% of the simulated ASR inputs. The next steps are to include the full year in every date the agent writes and to auto-fill the *order_id* during ticket creation so no handover is dropped.